Examiner only

2. *(a)* (i) State the products of protein digestion. [1]

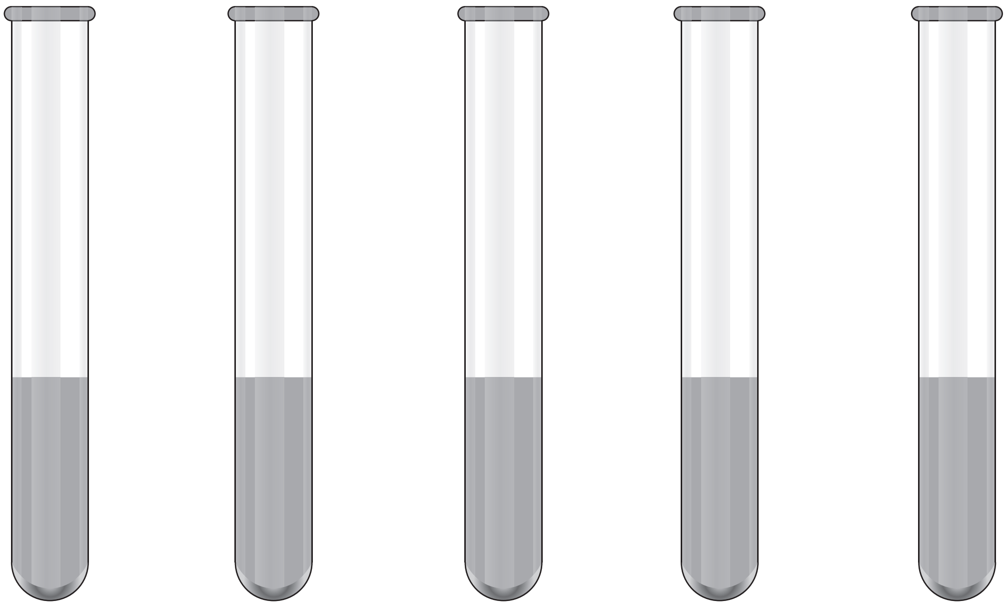
....

(ii) State the part of the digestive system that absorbs digested food molecules. [1]

....

(b) Protein digestion starts in the stomach.

Students investigated protein digestion. They set up five tubes **A** to **E**, as shown below.



Tube **A** **B** **C** **D** **E**

<table>
<tr><th>Tube A</th><th>Tube B</th><th>Tube C</th><th>Tube D</th><th>Tube E</th></tr>
<tr><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td></tr>
<tr><td>5 cm^3 of 0.1% protease</td><td rowspan="3">15 cm^3 liquid from the stomach</td><td>5 cm^3 of 0.1% protease</td><td rowspan="2">15 cm^3 distilled water</td><td rowspan="2">15 cm^3 distilled water</td></tr>
<tr><td>10 cm^3 distilled water</td><td>10 cm^3 distilled water</td></tr>
<tr><td>pH 2.0</td><td>pH 7.0</td><td>pH 2.0</td><td>pH 7.0</td></tr>
</table>

Examiner only

The students measured the percentage protein digested at one hour. The results are shown in the table.

Tube	Percentage protein digested
A	100
B	98
C	5
D	0
E	0

(i) Compare the results for tubes **A** and **D**. State the conclusion that can be made about the digestion of protein. [1]

....

....

(ii) State the conclusions that can be made about the contents of the liquid from the stomach in Tube **B**. [2]

....

....

....

....

(iii) All the 1% protein added to Tube **A** had been fully digested after one hour. State why the contents of Tube **A** would still test positive for protein. [1]

....

....

(iv) The students carried out a similar investigation, but instead of using 0.1% protease, they used 5 cm^3 of 0.1% lipase in both tubes **A** and **C**.
They found there had been no digestion of protein in either tube.
Use your knowledge of enzyme structure and function to explain the results that the students obtained. [2]

....

....

....

....

(v) State **one** further variable that should be controlled during this investigation. [1]

....

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